

MAKE CONNECTIONS

The Working Ecosystem

This arctic tundra ecosystem teems with life during the short two-month growing season each summer. In ecosystems, organisms interact with each other and with the environment around them in diverse ways, including those illustrated here.

Populations Are Dynamic (Chapter 53)

- 1 Caribou migrate across the tundra to give birth at their calving grounds each year. In general, populations change in size through births and deaths and through immigration and emigration. (See Figure 53.1 and Concept 53.1.)
- 2 Snow geese and many other species migrate to the Arctic each spring for the abundant food found there in summer. (See Concept 53.5.)
- 3 Birth and death rates influence the density of all populations. Death in the tundra comes from many causes, including predation, competition for resources, and lack of food in winter. (See Figure 53.17.)
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Species Interact in Diverse Ways (Chapter 54)

- 4 In predation, an individual of one species kills and eats another. (See Concept 54.1.)
- 5 In herbivory, an individual of one species eats part of a plant or other primary producer, such as a caribou eating a lichen. (See Concept 54.1.)
- 6 In mutualism, individuals of two species interact in ways that benefit each other. In some mutualisms, the partners live in direct contact, forming a symbiosis; for example, a lichen is a symbiotic mutualism between a fungus and an alga or cyanobacterium. (See Concept 54.1 and Figures 31.22 and 31.23.)
- 7 In competition, individuals of two species seek to acquire the same limiting resources. For example, snow geese and caribou both eat cottongrass. (See Concept 54.1.)